

Connection

Estimator base URL

http://localhost:8000

ESTIMATE_API_KEY

.....

Idempotency key (optional)

Slow + timeout caveat.

Generation uses gpt-5 with reasoning_effort=high — it takes minutes. The service default LLM_TIMEOUT=30 will 502. Raise it in estimator/.env (LLM_TIMEOUT=600) and recreate the container:

docker compose up -d --force-recreate estimator

Grounded Estimate — from a transcript

Session 9-11 flow · `POST /v1/estimate/from-transcript` · engineer-day breakdown with verifiable per-line citations.

Meeting transcript

[Kickoff call — NeoBank mobile application]

Client (CTO): We're building a mobile banking application for the Spanish market. We need OAuth 2.0 authentication with refresh-token rotation, and full PSD2 strong customer authentication (SCA) for every payment initiation.

PM (Ana): Understood. You also mentioned an immutable transaction ledger?

Client (CTO): Yes — a double-entry transaction ledger as the source of truth for balances, with an audit trail and nightly reconciliation jobs.

Backend Lead: We'll also need open banking connectors (AIS/PIS) plus token revocation and introspection endpoints and integration tests for the

Generate grounded estimate

X-Request-ID: 2f415dde-27b6-49a2-a2d6-be84f49d10f9

Total engineer-days

420

Duration (weeks)

—

Confidence

 medium

Reasoning (how the estimate was derived)

This estimate maps the requested scope to three closely matching historical components: OAuth 2.0 backend (120h), PSD2 AIS/PIS with SCA and consent (160h), and a double-entry ledger with audit trail and daily reconciliation (140h). Each component's hours are decomposed into concrete tasks that collectively sum to the source estimate, with per-line citations carrying verbatim evidence. Token revocation/introspection and integration tests for auth flows are included within the OAuth 2.0 backend allocation as they fall under the same component. We excluded unrelated sources (e.g., notifications, merchant dashboard, large-scale legacy ledger migration) to avoid inflating scope. Confidence is medium because the sources strongly align to the requested features, but specific sub-features (e.g., explicit mention of revocation/introspection endpoints, exact test coverage) are inferred within the cited OAuth baseline rather than enumerated explicitly in the sources.

Citation verification

Lines

20

Grounded

20

Insufficient

0

Verified citations

20

Every grounded line cites at least one retrieved chunk with verbatim evidence (shown under each task). Insufficient lines carry no invented hours. Dangling citations (ids never retrieved) are rejected server-side by the endpoint's corrective retry, so they are 0 by construction here.

Breakdown — modules → tasks (engineer-days)

📁 Authentication & Access (OAuth 2.0) · 120 engineer-days

OAuth 2.0 backend with refresh-token rotation, JWT sessions, multi-tenant isolation, client rate limiting, token revocation and introspection, plus auth-flow tests.

- 24d · Authorization Code + PKCE flow endpoints — ✅ grounded

Implement OAuth 2.0 authorization code flow with PKCE, authorization and token endpoints.

 chunk 1 · BUD-2024-001 — Component: OAuth 2.0 authentication backend Description: Implementation of OAuth 2.0 flows (authorization code, refresh token) with JWT-based session management, multi-tenant token isolation, and rate limiting per client. Estimated hours: 120

- 16d · Refresh-token rotation — ✅ grounded

Implement refresh-token issuance and rotation semantics.

 chunk 1 · BUD-2024-001 — Component: OAuth 2.0 authentication backend Description: Implementation of OAuth 2.0 flows (authorization code, refresh token) with JWT-based session management, multi-tenant token isolation, and rate limiting per client. Estimated hours: 120

- 16d · JWT-based session management — ✅ grounded

Configure JWT signing/verification, claims, and expiry handling.

 chunk 1 · BUD-2024-001 — Component: OAuth 2.0 authentication backend Description: Implementation of OAuth 2.0 flows (authorization code, refresh token) with JWT-based session management, multi-tenant token isolation, and rate limiting per client. Estimated hours: 120

- 16d · Multi-tenant token isolation — ✅ grounded

Enforce tenant scoping and isolation on token issuance and validation.

 chunk 1 · BUD-2024-001 — Component: OAuth 2.0 authentication backend Description: Implementation of OAuth 2.0 flows (authorization code, refresh token) with JWT-based session management, multi-tenant token isolation, and rate limiting per client. Estimated hours: 120

- 12d · Token revocation endpoint — ✅ grounded

Expose OAuth 2.0 token revocation endpoint and invalidate tokens server-side.

 chunk 1 · BUD-2024-001 — Component: OAuth 2.0 authentication backend Description: Implementation of OAuth 2.0 flows (authorization code, refresh token) with JWT-based session management, multi-tenant token isolation, and rate limiting per client. Estimated hours: 120

- 12d · Token introspection endpoint — ✅ grounded

Expose OAuth 2.0 token introspection endpoint for resource servers.

 chunk 1 · BUD-2024-001 — Component: OAuth 2.0 authentication backend Description: Implementation of OAuth 2.0 flows (authorization code, refresh token) with JWT-based session management, multi-tenant token isolation, and rate limiting per client. Estimated hours: 120

- 12d · Client rate limiting — ✅ grounded

Implement per-client rate limiting on auth endpoints.

 [chunk 1](#) · [BUD-2024-001](#) — Component: OAuth 2.0 authentication backend Description: Implementation of OAuth 2.0 flows (authorization code, refresh token) with JWT-based session management, multi-tenant token isolation, and rate limiting per client. Estimated hours: 120

- 12d · Integration and security tests for auth flows —  grounded

End-to-end and negative-path tests for OAuth flows, token rotation, and isolation.

 [chunk 1](#) · [BUD-2024-001](#) — Component: OAuth 2.0 authentication backend Description: Implementation of OAuth 2.0 flows (authorization code, refresh token) with JWT-based session management, multi-tenant token isolation, and rate limiting per client. Estimated hours: 120

▼ PSD2 Open Banking Connectors (AIS/PIS + SCA) · 160 engineer-days

Account Information Services (AIS) and Payment Initiation Services (PIS) APIs with SCA and consent management per PSD2/Berlin Group.

- 40d · AIS endpoints —  grounded

Implement account information endpoints (accounts, balances, transactions) per Berlin Group.

 [chunk 2](#) · [BUD-2024-001](#) — Component: PSD2 open banking connectors Description: Account information and payment initiation endpoints compliant with PSD2/Berlin Group, including strong customer authentication (SCA) and consent management. Estimated hours: 160

- 40d · PIS payment initiation endpoints —  grounded

Implement payment initiation endpoints (initiate payment, status, cancel) per Berlin Group.

 [chunk 2](#) · [BUD-2024-001](#) — Component: PSD2 open banking connectors Description: Account information and payment initiation endpoints compliant with PSD2/Berlin Group, including strong customer authentication (SCA) and consent management. Estimated hours: 160

- 30d · SCA integration for payment initiation —  grounded

Enforce PSD2 SCA for all PIS flows, linking auth and connector layers.

 [chunk 2](#) · [BUD-2024-001](#) — Component: PSD2 open banking connectors Description: Account information and payment initiation endpoints compliant with PSD2/Berlin Group, including strong customer authentication (SCA) and consent management. Estimated hours: 160

- 25d · Consent management —  grounded

Implement consent grant, scope, lifecycle (create, list, revoke) for AIS/PIS.

 [chunk 2](#) · [BUD-2024-001](#) — Component: PSD2 open banking connectors Description: Account information and payment initiation endpoints compliant with PSD2/Berlin Group, including strong customer authentication (SCA) and consent management. Estimated hours: 160

- 15d · Berlin Group compliance mapping —  grounded

Profile mapping, error codes, and conformance alignment to PSD2/Berlin Group.

 [chunk 2](#) · [BUD-2024-001](#) — Component: PSD2 open banking connectors Description: Account information and payment initiation endpoints compliant with PSD2/Berlin Group, including strong customer authentication (SCA) and consent management. Estimated hours: 160

- 10d · AIS/PIS integration test harness —  grounded

Build integration test harness and stubs for AIS/PIS flows.

 [chunk 2](#) · [BUD-2024-001](#) — Component: PSD2 open banking connectors Description: Account information and payment initiation endpoints compliant with PSD2/Berlin Group, including strong customer authentication (SCA) and consent management. Estimated hours: 160

▼ Double-entry Ledger & Nightly Reconciliation · 140 engineer-days

Immutable, double-entry transaction ledger with idempotent writes, audit trail and nightly reconciliation jobs.

- 24d · Ledger schema and account model —  grounded

Design chart of accounts, entries, postings, and balance computation schema.

 [chunk 3](#) · [BUD-2024-001](#) — Component: Transaction ledger service Description: Double-entry transaction ledger with idempotent writes, daily reconciliation jobs and an audit trail of every balance change. Estimated hours: 140

- 36d · Posting engine with idempotent writes —  grounded

Implement double-entry posting logic with idempotency keys and dedupe.

 [chunk 3](#) · [BUD-2024-001](#) — Component: Transaction ledger service Description: Double-entry transaction ledger with idempotent writes, daily reconciliation jobs and an audit trail of every balance change. Estimated hours: 140

- 24d · Immutable audit trail —  grounded

Record and expose an audit trail for every balance change and posting.

 [chunk 3](#) · [BUD-2024-001](#) — Component: Transaction ledger service Description: Double-entry transaction ledger with idempotent writes, daily reconciliation jobs and an audit trail of every balance change. Estimated hours: 140

- 28d · Nightly reconciliation jobs —  grounded

Implement daily/nightly reconciliation jobs and discrepancy reporting.

 [chunk 3](#) · [BUD-2024-001](#) — Component: Transaction ledger service Description: Double-entry transaction ledger with idempotent writes, daily reconciliation jobs and an audit trail of every balance change. Estimated hours: 140

- 12d · Idempotency boundary and keys —  grounded

Define idempotency key scheme and deduplication window for postings.

 [chunk 3](#) · [BUD-2024-001](#) — Component: Transaction ledger service Description: Double-entry transaction ledger with idempotent writes, daily reconciliation jobs and an audit trail of every balance change. Estimated hours: 140

- 16d · Ledger APIs (balances/statements) —  grounded

Expose read APIs for balances and statements backed by ledger.

 [chunk 3](#) · [BUD-2024-001](#) — Component: Transaction ledger service Description: Double-entry transaction ledger with idempotent writes, daily reconciliation jobs and an audit trail of every balance change. Estimated hours: 140

Assumptions (scope without a historical analog)	Estimate-global sources
<i>high</i> — Bank of Spain supervised launch activities (policies, risk/ops controls, external audits, compliance documentation, approvals) are out of scope of the cited build estimates. No provided source quantifies regulatory launch-readiness overhead; only engineering components are budgeted. <i>high</i> — Production-grade infrastructure/operations (secrets/HSM, monitoring/alerting, logging/ SIEM, backups/DR, runbooks) not included. None of the sources include infra/devops hours; adding them would require non-sourced estimation. <i>medium</i> — Mobile app UX, device enrollment, and customer-facing SCA UX flows are excluded; backend/API only. Sources focus on backend APIs and services, not client applications. <i>high</i> — Third-party penetration testing and PSD2 certification/conformance testing by external labs are excluded. No source includes external testing/certification budgets. <i>medium</i> — Data migration from any legacy core or ledger is not included. The relevant migration source is about a strangler migration, which is not applicable to this greenfield scope and is not used in this estimate.	<i>chunk 1</i> · primary — OAuth 2.0 backend module breakdown and task effort distribution based on 120h estimate. <i>chunk 2</i> · primary — PSD2 open banking connectors (AIS/PIS, SCA, consent) module breakdown and effort distribution from 160h estimate. <i>chunk 3</i> · primary — Double-entry ledger with idempotent writes, audit trail and daily reconciliation module, distributed from 140h estimate.

Raw Estimate JSON

```
{
  "total_engineer_days": 420
  "modules": [
    0: {
      "name": "Authentication & Access (OAuth 2.0)"
      "description": "OAuth 2.0 backend with refresh-token rotation, JWT sessions, multi-tenant isolation, client rate limiting, token revocation and introspection, plus auth-flow tests."
      "tasks": [
        0: {
          "name": "Authorization Code + PKCE flow endpoints"
          "description": "Implement OAuth 2.0 authorization code flow with PKCE, authorization and token endpoints."
          "engineer_days": 24
          "grounded": true
          "sources": [
            0: {
              "chunk_id": "1"
              "document_id": "BUD-2024-001"
            }
          ]
        }
      ]
    }
  ]
}
```

```
      "evidence" :
      "Component: OAuth 2.0 authentication backend\nDescription: Implementation of OAuth 2.0 flows (authorization code, refresh token) with JWT-based session management, multi-tenant token isolation, and rate limiting per client.\nEstimated hours: 120"
    }
  ]
}

1 : {
  "name" : "Refresh-token rotation"
  "description" :
  "Implement refresh-token issuance and rotation semantics."
  "engineer_days" : 16
  "grounded" : true
  "sources" : [
    0 : {
      "chunk_id" : "1"
      "document_id" : "BUD-2024-001"
      "evidence" :
      "Component: OAuth 2.0 authentication backend\nDescription: Implementation of OAuth 2.0 flows (authorization code, refresh token) with JWT-based session management, multi-tenant token isolation, and rate limiting per client.\nEstimated hours: 120"
    }
  ]
}

2 : {
  "name" : "JWT-based session management"
  "description" :
  "Configure JWT signing/verification, claims, and expiry handling."
  "engineer_days" : 16
  "grounded" : true
  "sources" : [
    0 : {
      "chunk_id" : "1"
      "document_id" : "BUD-2024-001"
      "evidence" :
      "Component: OAuth 2.0 authentication backend\nDescription: Implementation of OAuth 2.0 flows (authorization code, refresh token) with JWT-based session management, multi-tenant token isolation, and rate limiting per client.\nEstimated hours: 120"
    }
  ]
}

3 : {
  "name" : "Multi-tenant token isolation"
  "description" :
  "Enforce tenant scoping and isolation on token issuance and validation."
  "engineer_days" : 16
  "grounded" : true
  "sources" : [
    0 : {
      "chunk_id" : "1"
```

```
"document_id" : "BUD-2024-001"
"evidence" :
"Component: OAuth 2.0 authentication backend\nDescription: Implementation of OAuth 2.0 flows (authorization code, refresh token) with JWT-based session management, multi-tenant token isolation, and rate limiting per client.\nEstimated hours: 120"
}
]
}
4 : {
  "name" : "Token revocation endpoint"
  "description" :
  "Expose OAuth 2.0 token revocation endpoint and invalidate tokens server-side."
  "engineer_days" : 12
  "grounded" : true
  "sources" : [
    0 : {
      "chunk_id" : "1"
      "document_id" : "BUD-2024-001"
      "evidence" :
      "Component: OAuth 2.0 authentication backend\nDescription: Implementation of OAuth 2.0 flows (authorization code, refresh token) with JWT-based session management, multi-tenant token isolation, and rate limiting per client.\nEstimated hours: 120"
    }
  ]
}
5 : {
  "name" : "Token introspection endpoint"
  "description" :
  "Expose OAuth 2.0 token introspection endpoint for resource servers."
  "engineer_days" : 12
  "grounded" : true
  "sources" : [
    0 : {
      "chunk_id" : "1"
      "document_id" : "BUD-2024-001"
      "evidence" :
      "Component: OAuth 2.0 authentication backend\nDescription: Implementation of OAuth 2.0 flows (authorization code, refresh token) with JWT-based session management, multi-tenant token isolation, and rate limiting per client.\nEstimated hours: 120"
    }
  ]
}
6 : {
  "name" : "Client rate limiting"
  "description" :
  "Implement per-client rate limiting on auth endpoints."
  "engineer_days" : 12
  "grounded" : true
  "sources" : [
```

```
    0 : {
      "chunk_id" : "1"
      "document_id" : "BUD-2024-001"
      "evidence" :
        "Component: OAuth 2.0 authentication backend\nDescription: Implementation of OAuth 2.0 flows (authorization code, refresh token) with JWT-based session management, multi-tenant token isolation, and rate limiting per client.\nEstimated hours: 120"
    }
  ]
}

7 : {
  "name" : "Integration and security tests for auth flows"
  "description" :
    "End-to-end and negative-path tests for OAuth flows, token rotation, and isolation."
  "engineer_days" : 12
  "grounded" : true
  "sources" : [
    0 : {
      "chunk_id" : "1"
      "document_id" : "BUD-2024-001"
      "evidence" :
        "Component: OAuth 2.0 authentication backend\nDescription: Implementation of OAuth 2.0 flows (authorization code, refresh token) with JWT-based session management, multi-tenant token isolation, and rate limiting per client.\nEstimated hours: 120"
    }
  ]
}

1 : {
  "name" : "PSD2 Open Banking Connectors (AIS/PIS + SCA)"
  "description" :
    "Account Information Services (AIS) and Payment Initiation Services (PIS) APIs with SCA and consent management per PSD2/Berlin Group."
  "tasks" : [
    0 : {
      "name" : "AIS endpoints"
      "description" :
        "Implement account information endpoints (accounts, balances, transactions) per Berlin Group."
      "engineer_days" : 40
      "grounded" : true
      "sources" : [
        0 : {
          "chunk_id" : "2"
          "document_id" : "BUD-2024-001"
          "evidence" :
            "Component: PSD2 open banking connectors\nDescription: Account information and payment initiation endpoints compliant with PSD2/Berlin Group, including strong customer authentication (SCA) and consent management.\nEstimated hours: 160"
        }
      ]
    }
  ]
}
```



```
}
  1: {
    "name": "PIS payment initiation endpoints"
    "description":
      "Implement payment initiation endpoints (initiate payment, status, cancel) per Berlin Group."
    "engineer_days": 40
    "grounded": true
    "sources": [
      0: {
        "chunk_id": "2"
        "document_id": "BUD-2024-001"
        "evidence":
          "Component: PSD2 open banking connectors\nDescription: Account information and payment initiation endpoints compliant with PSD2/Berlin Group, including strong customer authentication (SCA) and consent management.\nEstimated hours: 160"
      }
    ]
  }
  2: {
    "name": "SCA integration for payment initiation"
    "description":
      "Enforce PSD2 SCA for all PIS flows, linking auth and connector layers."
    "engineer_days": 30
    "grounded": true
    "sources": [
      0: {
        "chunk_id": "2"
        "document_id": "BUD-2024-001"
        "evidence":
          "Component: PSD2 open banking connectors\nDescription: Account information and payment initiation endpoints compliant with PSD2/Berlin Group, including strong customer authentication (SCA) and consent management.\nEstimated hours: 160"
      }
    ]
  }
  3: {
    "name": "Consent management"
    "description":
      "Implement consent grant, scope, lifecycle (create, list, revoke) for AIS/PIS."
    "engineer_days": 25
    "grounded": true
    "sources": [
      0: {
        "chunk_id": "2"
        "document_id": "BUD-2024-001"
        "evidence":
          "Component: PSD2 open banking connectors\nDescription: Account information and payment initiation endpoints compliant with PSD2/Berlin Group, including strong customer authentication (SCA) and consent management.\nEstimated hours: 160"
      }
    ]
  }
}
```

```
}
  4: {
    "name": "Berlin Group compliance mapping"
    "description":
      "Profile mapping, error codes, and conformance alignment to PSD2/Berlin Group."
    "engineer_days": 15
    "grounded": true
    "sources": [
      0: {
        "chunk_id": "2"
        "document_id": "BUD-2024-001"
        "evidence":
          "Component: PSD2 open banking connectors\nDescription: Account information and payment initiation endpoints compliant with PSD2/Berlin Group, including strong customer authentication (SCA) and consent management.\nEstimated hours: 160"
      }
    ]
  }
  5: {
    "name": "AIS/PIS integration test harness"
    "description":
      "Build integration test harness and stubs for AIS/PIS flows."
    "engineer_days": 10
    "grounded": true
    "sources": [
      0: {
        "chunk_id": "2"
        "document_id": "BUD-2024-001"
        "evidence":
          "Component: PSD2 open banking connectors\nDescription: Account information and payment initiation endpoints compliant with PSD2/Berlin Group, including strong customer authentication (SCA) and consent management.\nEstimated hours: 160"
      }
    ]
  }
]
}

2: {
  "name": "Double-entry Ledger & Nightly Reconciliation"
  "description":
    "Immutable, double-entry transaction ledger with idempotent writes, audit trail and nightly reconciliation jobs."
  "tasks": [
    0: {
      "name": "Ledger schema and account model"
      "description":
        "Design chart of accounts, entries, postings, and balance computation schema."
      "engineer_days": 24
      "grounded": true
      "sources": [
        0: {
          "chunk_id": "3"
          "document_id": "BUD-2024-001"
```

```
      "evidence" :  
      "Component: Transaction ledger service\nDescription: Double-entry transaction ledger with idempotent writes, daily reconciliation jobs and an audit trail of every balance change.\nEstimated hours: 140"  
    }  
  ]  
}  
  
1 : {  
  "name" : "Posting engine with idempotent writes"  
  "description" :  
  "Implement double-entry posting logic with idempotency keys and dedupe."  
  "engineer_days" : 36  
  "grounded" : true  
  "sources" : [  
    0 : {  
      "chunk_id" : "3"  
      "document_id" : "BUD-2024-001"  
      "evidence" :  
      "Component: Transaction ledger service\nDescription: Double-entry transaction ledger with idempotent writes, daily reconciliation jobs and an audit trail of every balance change.\nEstimated hours: 140"  
    }  
  ]  
}  
  
2 : {  
  "name" : "Immutable audit trail"  
  "description" :  
  "Record and expose an audit trail for every balance change and posting."  
  "engineer_days" : 24  
  "grounded" : true  
  "sources" : [  
    0 : {  
      "chunk_id" : "3"  
      "document_id" : "BUD-2024-001"  
      "evidence" :  
      "Component: Transaction ledger service\nDescription: Double-entry transaction ledger with idempotent writes, daily reconciliation jobs and an audit trail of every balance change.\nEstimated hours: 140"  
    }  
  ]  
}  
  
3 : {  
  "name" : "Nightly reconciliation jobs"  
  "description" :  
  "Implement daily/nightly reconciliation jobs and discrepancy reporting."  
  "engineer_days" : 28  
  "grounded" : true  
  "sources" : [  
    0 : {  
      "chunk_id" : "3"  
      "document_id" : "BUD-2024-001"
```

```
      "evidence" :  
      "Component: Transaction ledger service\nDescription: Double-entry transaction ledger with idempotent writes, daily reconciliation jobs and an audit trail of every balance change.\nEstimated hours: 140"  
    }  
  ]  
}  
  
4 : {  
  "name" : "Idempotency boundary and keys"  
  "description" :  
  "Define idempotency key scheme and deduplication window for posting s."  
  "engineer_days" : 12  
  "grounded" : true  
  "sources" : [  
    0 : {  
      "chunk_id" : "3"  
      "document_id" : "BUD-2024-001"  
      "evidence" :  
      "Component: Transaction ledger service\nDescription: Double-entry transaction ledger with idempotent writes, daily reconciliation jobs and an audit trail of every balance change.\nEstimated hours: 140"  
    }  
  ]  
}  
  
5 : {  
  "name" : "Ledger APIs (balances/statements)"  
  "description" :  
  "Expose read APIs for balances and statements backed by ledger."  
  "engineer_days" : 16  
  "grounded" : true  
  "sources" : [  
    0 : {  
      "chunk_id" : "3"  
      "document_id" : "BUD-2024-001"  
      "evidence" :  
      "Component: Transaction ledger service\nDescription: Double-entry transaction ledger with idempotent writes, daily reconciliation jobs and an audit trail of every balance change.\nEstimated hours: 140"  
    }  
  ]  
}  
]  
}  
  
]  
  
"duration_weeks" : NULL  
"sources" : [  
  0 : {  
    "source_id" : 1  
    "relevance" : "primary"  
    "used_for" :  
    "OAuth 2.0 backend module breakdown and task effort distribution based on 120h estimate."  
  }  
]
```

```
  1 : {
    "source_id" : 2
    "relevance" : "primary"
    "used_for" :
      "PSD2 open banking connectors (AIS/PIS, SCA, consent) module breakdown and effort distribution from 160h estimate."
  }
  2 : {
    "source_id" : 3
    "relevance" : "primary"
    "used_for" :
      "Double-entry ledger with idempotent writes, audit trail and daily reconciliation module, distributed from 140h estimate."
  }
]
"assumptions" : [
  0 : {
    "description" :
      "Bank of Spain supervised launch activities (policies, risk/ops controls, external audits, compliance documentation, approvals) are out of scope of the cited build estimates."
    "impact" : "high"
    "rationale" :
      "No provided source quantifies regulatory launch-readiness overhead; only engineering components are budgeted."
  }
  1 : {
    "description" :
      "Production-grade infrastructure/operations (secrets/HSM, monitoring/alerting, logging/ SIEM, backups/DR, runbooks) not included."
    "impact" : "high"
    "rationale" :
      "None of the sources include infra/devops hours; adding them would require on-sourced estimation."
  }
  2 : {
    "description" :
      "Mobile app UX, device enrollment, and customer-facing SCA UX flows are excluded; backend/API only."
    "impact" : "medium"
    "rationale" :
      "Sources focus on backend APIs and services, not client applications."
  }
  3 : {
    "description" :
      "Third-party penetration testing and PSD2 certification/conformance testing by external labs are excluded."
    "impact" : "high"
    "rationale" : "No source includes external testing/certification budgets."
  }
  4 : {
    "description" :
      "Data migration from any legacy core or ledger is not included."
    "impact" : "medium"
    "rationale" :
      "The relevant migration source is about a strangler migration, which is not applicable to this greenfield scope and is not used in this estimate."
  }
]
```

```
}  
]  
"confidence" : "medium"  
"reasoning" :  
"This estimate maps the requested scope to three closely matching historical components: OAuth 2.0 backend (120h), PSD2 AIS/PIS with SCA and consent (160h), and a double-entry ledger with audit trail and daily reconciliation (140h). Each component's hours are decomposed into concrete tasks that collectively sum to the source estimate, with per-line citations carrying verbatim evidence. Token revocation/introspection and integration tests for auth flows are included within the OAuth 2.0 backend allocation as they fall under the same component. We excluded unrelated sources (e.g., notifications, merchant dashboard, large-scale legacy ledger migration) to avoid inflating scope. Confidence is medium because the sources strongly align to the requested features, but specific sub-features (e.g., explicit mention of revocation/introspection endpoints, exact test coverage) are inferred within the cited OAuth baseline rather than enumerated explicitly in the sources."  
"insufficient_context_explanation" : NULL  
}
```